Runci Bai

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EDUCATION

China Academy of Telecommunications Technology (CATT) **Beijing, China**
Master of Engineering in Information and Communication Engineering Sep. 2024 - May 2027 (Expected)
China University of Mining and Technology (CUMT) **Xuzhou, China**
Bachelor of Engineering in Automation Sep. 2020 - Jun. 2024

PUBLICATIONS/UNDER REVIEW

Runci Bai, et al. "SCC-YOLO: An Improved Object Detector for Assisting in Brain Tumor Diagnosis." HBD 25: Proceedings of the 2025 International Conference on Health Big Data, May 2025. DOI: 10.1145/3733006.3733026.

Runci Bai, et al. "Loop-Mamba: A Loop Mamba with Degradation-Aware and Shared Memory for Old Photo Restoration." Submitted to the AAAI Conference on Artificial Intelligence (**AAAI**), arXiv:2608.02346.

Runci Bai, et al. "6thGrid-Net: Unified Remote Sensing Image Dehazing Based on Color Restoration and Edge-Preserving." Submitted to IEEE Transactions on Image Processing **(IEEE TIP**), arXiv:2604.24149.

Runci Bai, et al. "SMR-SAM: Joint Modeling of Cross-Modal Semantic Interactions and Boundary-Aware Representations for Brain Tumor Segmentation." Submitted to the IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**).

Yucheng Xin, **Runci Bai**, Yongcong Wang, et al. "LoopMoEVR: Loop-Based Degradation-Aware Mixture-of-Experts for Unified UHD Video Restoration." Submitted to the International Conference on Learning Representations (**ICLR**).

Yongcong Wang, Pu Wang, Hingchin Chen, **Runci Bai**, et al. "MoCRA: Mixture of Compositional Rank-1 Atoms for 4K All-in-One Video Restoration." Submitted to the AAAI Conference on Artificial Intelligence (**AAAI**), arXiv:2608.01829

Mingzhe Liu, **Runci Bai**, Jia Wang. "RESOFT: Reliable Soft Supervision for Two-Stage Semi-Supervised Object Detection." Submitted to the IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**).

Huixiang Xiao, Zhengyang Han, **Runci Bai**, et al. "Med-LPM: Balanced Latent Subspace Parameterization for Medical Image Classification." Submitted to the IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**).

Jiahui Qu, Ziqiong Li, Yifang Qin, **Runci Bai**, et al. "WorldCut: Certified Self-Interruption for Spoken Agents When the World Changes Mid-Sentence." Submitted to the IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**).

XU Guibao, **BAI Runci**, YI Hongjie. Research on construction of an integrated service ecosystem for medical, health, and elderly care empowered by AI[J]. Information and Communications Technology and Policy, 2025, 51(8): 84-90.

Runci Bai, Kai Wang, Fanwen Yu. "A ResNet-VD Based Intelligent Garbage Sorting System." Environmental Engineering, Nov. 2023.

RESEARCH EXPERIENCE

SMR-SAM: A Joint Multimodal Fusion and Boundary-Aware Learning Framework for Brain Tumor Segmentation **Beijing, China**
Team Lead; Advisor: Guibao Xu, Senior Engineer, CATT Nov. 2025 - Jul. 2026

- Designed a SAM-based segmentation framework for multimodal MRI, integrating T1, T2, T1ce, and FLAIR inputs through a four-modality fusion module (FFM) and boundary enhancement module (BEM).
- Designed a hybrid boundary-aware optimization strategy that combines Dice Loss, Hausdorff Distance Loss, and Boundary-aware Dice Loss to improve geometric consistency on challenging tumor boundaries.
- Prepared the BraTS 2021 and BraTS 2023-SSA datasets and performed ablation experiments to validate the effectiveness of FFM and BEM and analyze the impact of different modality combinations on segmentation performance.
- Achieved an mDice of 0.85, an mIoU of 0.74, and an HD95 of 8.73 on BraTS 2021, as well as an mDice of 0.70, an mIoU of 0.56, and an HD95 of 14.32 on BraTS 2023-SSA.

BiCycle: Towards Modality-Unified Visible-Infrared Person Re-Identification via Identity-Preserving Semantic Consensus **Beijing, China**

Second Author; Advisor: Yongfei Zhang, Professor, Beihang University Nov. 2025 - Apr. 2026

- Developed a modality-unification pipeline that uses grayscale images as a semantic bridge between visible and infrared person images.
- Contributed to the design and construction of the identity-preserving reconstruction, identity-center clustering, and attribute-distribution distillation modules; validated module contributions through ablation experiments.
- Evaluated the model on the SYSU-MM01, RegDB, and LLCM datasets and conducted comparative analyses.
- Achieved 69.84% Rank-1 / 67.24% mAP under the All-Search mode, and 76.71% Rank-1 / 80.58% mAP under the Indoor-Search mode of the SYSU-MM01 dataset; reported consistent Rank-1 gains of 0.8-2.8 points when used as a plug-in preprocessing module.

6thGrid-Net: Unified Remote Sensing Image Dehazing Based on Color Restoration and Edge-Preserving **Beijing, China**

Independent Researcher; Advisor: Zhuoran Zheng, Assoc. Prof., Qilu Univ. of Technology Mar. 2026 - Apr. 2026

- Designed a lightweight restoration framework that integrates 3D LUT color recovery with bilateral-grid spatial and luminance detail preservation through a six-dimensional fusion tensor I constructed.
- Developed manifold-adaptive high-dimensional sampling to adjust interpolation kernels by local edge direction, texture strength, and color similarity, reducing cross-edge blur and halo artifacts.
- Built an edge-aware grid smoothness constraint and dynamic INT8 quantization pipeline, compressing the model from 23.48 MB to 7.84 MB and reducing FLOPs from 42.0G to 8.09G.
- Evaluated the model on 1,700 paired remote sensing images and the ViCoW colorization dataset, achieving PSNR of 22.52 dB, SSIM of 0.836, and LPIPS of 0.164 for remote sensing images, and PSNR of 28.61 dB, SSIM of 0.965, and LPIPS of 0.133 for the ViCoW dataset, respectively.

Loop-Mamba: A Loop Mamba with Degradation-Aware and Shared Memory for OldPhoto Restoration **Beijing, China**

Independent Researcher; Advisor: Zhuoran Zheng, Assoc. Prof., Qilu Univ. of Technology Mar. 2026 - Apr. 2026

- Proposed and designed the framework combining semantic supervision, directional scanning, and shared memory for complex old-photo degradation patterns.
- Built a CLIP-guided degradation estimator to learn pixel-level degradation maps and global degradation scores.
- Designed a direction prediction module that learns horizontal, vertical, diagonal, and anti-diagonal scan weights to improve texture continuity and edge restoration.
- Implemented shared memory across scan cycles to accumulate historical features and strengthen long-range context modeling for large missing or severely degraded regions.
- Conducted ablation experiments, lightweight analysis, and performance evaluation to verify each module's contribution to old photo restoration.

BE-SAM: A Dual-Branch Boundary-Enhanced Foundation Model for Breast Ultrasound Tumor Segmentation **Beijing, China**

Independent Researcher; Advisor: Guibao Xu, Senior Engineer, CATT Jul. 2025 - Jan. 2026

- Embedded dual-branch boundary enhancement modules into the SAM mask decoder, using edge convolution and feature convolution branches to sharpen tumor boundaries after upsampling.
- Trained and evaluated the model on the Breast Ultrasound Images Dataset (BUSI) and BUS-synthetic datasets with mDice, mIoU, and HD95 metrics.
- Performed ablation experiments to determine the optimal placement of the module.
- Achieved an mDice of 0.9083, an mIoU of 0.8339, and an HD95 of 36.4542 on BUSI, substantially improving over the baseline SAM and validating external generalization on BUS-synthetic.

SCC-YOLO: An Improved Object Detector for Assisting in Brain Tumor Diagnosis **Beijing, China**

Project Lead; Advisor: Guibao Xu, Senior Engineer, CATT Oct. 2024 - Mar. 2025

- Integrated the SCConv attention mechanism into YOLOv9 to strengthen spatial and channel feature extraction for brain tumor detection in MRI images.
- Prepared the Br35H dataset and a self-built dataset (Brain_Tumor_Dataset), which involved cleaning, labeling, and a 7,920/1,980 train-test split.
- Conducted YOLOv9c training on an RTX 3090 and compared the performance of SCConv with SE attention in brain tumor image detection under matched settings.
- Improved mAP50 to 0.957 on Br35H, a 0.3% improvement over YOLOv9, and reached 0.860 on the self-built dataset, a 0.5% improvement over YOLOv9, while reducing parameters compared to SE-YOLOv9.

An Everyday Object Recognition System Utilizing a Lightweight Object Detection Model **Xuzhou, China**
Final Year Project; Advisor: Meiqiang Zhu, Associate Professor, CUMT Mar. 2024 - Jun. 2024

- Utilized Raspberry Pi 4B and YOLOv7-tiny to develop a cost-effective everyday object recognition system.
- Constructed a dataset of 16,480 images and performed annotation and augmentation.
- Trained the model for 50 epochs using YOLOv7-tiny as the baseline and achieved optimal performance.
- Introduced MobileOne to optimize the ELAN structure of YOLOv7-tiny while combining EIOU loss and SGD optimizer to improve the recognition speed of the original model.
- Implemented a front-end display interface and voice broadcast function to visualize and interactively report detection results.

A ResNet-Vd-Based Intelligent Garbage Sorting System **Xuzhou, China**
Project Lead; Advisor: Zhihui Hu, CUMT Jun. 2021 - Jun. 2023

- Built a hybrid dataset using Huawei Cloud public data and self-collected images.
- Trained a ResNet-Vd model using PaddleClas with RandomErasing augmentation for 60 epochs.
- Converted the trained model to an inference format and deployed it to a Raspberry Pi 4B.
- Integrated a 720P monocular camera for waste image capture and realized precise tray rotation and dumping by controlling a dual-axis servo module.
- Developed a GUI using PyQt5 that displays promotional videos, waste category information, and real-time bin fullness status (detected by an HC-SR04 ultrasonic sensor).

PROFESSIONAL EXPERIENCE

China Academy of Information and Communications Technology **Beijing, China**
Digital Health Department Intern, Institute of Cloud Computing and Digitalization Jul. 2025 - Present

- Engaged in two national-level research projects: “Research on Construction of an Integrated Service Ecosystem for Medical, Health, and Elderly Care Empowered by AI” and “Research on Exercise and Health Data Circulation Technologies and Platform Development.”
- Contributed to on-site research and demand analysis for the AI planning and design project at Guizhou Provincial People’s Hospital by conducting structured interviews.
- Drafted 12 department-level research notes and an AI planning report covering data governance, algorithmic applications, and phased implementation.
- Proposed regionalized disease characteristics-driven AI application recommendations that were adopted by the project team.

Mindray Medical International (Beijing Research Center) **Beijing, China**
Algorithm and Application Development Engineer Intern Feb. 2025 - Jun. 2025

- Processed approximately 3,000 cardiac, breast, and thyroid ultrasound images and supported SAM fine-tuning with PyTorch for medical-image segmentation.
- Conducted model training and deployment tests on the NVIDIA Tesla T4 and a local medical inference device (Intel CPU + NVIDIA Jetson platform).
- Optimized inference speed through TensorRT and half-precision inference (FP16), reducing image segmentation time from 350 ms to under 120 ms.
- Completed 12 control experiments, achieving a Dice score of 0.91 for echocardiogram images, 0.88 for breast images, and 0.89 for thyroid images.
- Participated in writing the “SAM Model Ultrasound Image Deployment Guide.”

Jiangsu Noon Crop Science Co., Ltd. **Xuzhou, China**
Artificial Intelligence Data Annotation Intern Jul. 2023 - Aug. 2023

- Annotated 2,000+ industrial component images in LabelImg, with a labeling accuracy of over 98%, for a computer vision defect-detection dataset.
- Standardized boundary decisions, conducted quality checks, and corrected 120+ annotation deviations.
- Generated XML and TXT label files and organized them by component model and crack type.
- Created a crack-annotation guideline adopted for new team member onboarding.

TECHNICAL SKILLS

Programming: Python, C
Machine Learning: PyTorch, YOLO, SAM, ResNet, U-Net, Mamba-based vision models, medical image segmentation
Other Tools: MATLAB, AutoCAD, LabelImg, TensorRT, CUDA, Excel